Ear as an XPhone: What a Consumer EEG Headband Records of Sound Through Air, Water and Solid

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***Abstract*—The eye as a camera reads a picture from the visual cortex. The ear as an XPhone asks the same of hearing, with X the state of matter the sound travels in: through air the ear is a microphone, through water a hydrophone, the medium of the hydraulophone, and through a solid a geophone. We report what a consumer headband (Muse S, four dry electrodes, phone app) recorded in each. *Air:* a spoken passage played 24 times; a linear decoder read its loudness envelope back from the EEG at $r = 0.36$ (shuffled null 95th percentile 0.10, $p < 0.001$; control at chance), and an EEG-conditioned music generator built by a co-worker produced clips whose rhythm followed the heard one more closely than with shuffled EEG in 6 of 6 examples, although its semantic score was not yet significant. A steady-state test of a 40 Hz-modulated tone gave no response above a floor of 0.04 μV. *Water:* six minutes lying at the edge of a dock beside a hydraulophone, the head out of the water, gave clean EEG and a textbook optical pulse but no trace of the music, whose notes lie above the headband's 128 Hz Nyquist limit. Seven people at once then listened through a hydrophone to swimmers jumping into the lake, a kick and a stick striking the water, each wearing a headband recorded by one phone; their forehead EEG and head motion showed no moment shared beyond chance. *Solid:* four minutes of tapping and scratching a table, the head not touching it, moved the head by 2.8 to 5.6 mg on the headband's accelerometer through the arms and body, yet the EEG neither followed that motion (coherence at chance) nor showed a reliable tap-locked response. The ear is a microphone on a consumer headband; as a hydrophone and a geophone it needs a tagged stimulus, and we give the protocol.**

***Index Terms*—auditory EEG; speech envelope decoding; hydraulophone; underwater hearing; bone conduction; geophone; auditory steady-state response; wearable computing; consumer electronics; Mersivity.**

I. Introduction

Mann's physiphones are instruments named for the state of matter that makes their sound: the hydraulophone sounds in water, the pyrophone in fire, the ordinary instruments in air and in solids [1]–[3]. The ear as an XPhone turns that taxonomy on the listener. Sound is made in air, in water or in a solid, and reaches the ear by whichever path joins them: through air directly; from water or a solid either across the boundary into the air, which reflects most of the energy, or by bone conduction when the head touches the medium, with underwater thresholds some 30 to 60 dB above those in air and a response that favours the low kilohertz [4], [5]. In the recordings here the sound was made in each medium and the listener's head stayed in air, out of the water and off the table, so they test the first path.

Our eye-as-camera work read what the visual cortex does with a flickering target from a consumer headband [6]. Here we ask what the same headband records of sound in each medium. The brain follows the loudness envelope of speech, and a linear decoder can read it back from the scalp [7], [8]; it follows a sound's amplitude modulation near 40 Hz as the auditory steady-state response (ASSR) [9], [10]; and the headband also carries an accelerometer and an optical pulse, which record what the medium does to the head itself [11], [12]. We report one recording per medium, and say for each what the headband gave and what the next recording must add.

II. Recording

All recordings used a Muse S headband: four dry EEG electrodes at TP9, AF7, AF8 and TP10 referenced at Fpz, 256 Hz, with an accelerometer and gyroscope and (Muse S Athena) an optical array [13], streamed to a phone running our app MuseLog [14], which records every channel and can forward the EEG to a laptop. One listener per medium; no participant other than the authors.

III. Air: the Ear as a Microphone

Speech envelope. The first two paragraphs of *Alice's Adventures in Wonderland* (a public-domain LibriVox reading, 31.5 s) were played 24 times from the speakers of a Meta Quest 3S worn by the listener, the first author, who looked at a dot. A ridge decoder on the five channels (the four Muse electrodes and a cup on the scalp) at lags of -125 to $+500$ ms read the 1 to 8 Hz envelope; each play was decoded by a model trained on the others, its first and last 3 s left out, and the mean of the held-out reconstructions is the brain's envelope of the clip (Fig. 1). With every electrode in contact it correlated with the true envelope at $r = 0.36$ over 23 plays (circular-shift null 95th percentile 0.10, $p < 0.001$), one play at 0.086 (null 0.022), and with the EEG shifted off the clips at chance (0.035, $p = 0.29$); a night session with the cup off gave 0.18 ($p < 0.001$; control 0.078). The signal sits on the ear electrodes and the cup (TP10 0.20, AF8 0.16, cup 0.15, TP9 0.14 alone). Research caps with 64 channels reach 0.10 to 0.19 for one play of continuous speech [8]; repetition gives the headband what it lacks in channels. The headset's speakers

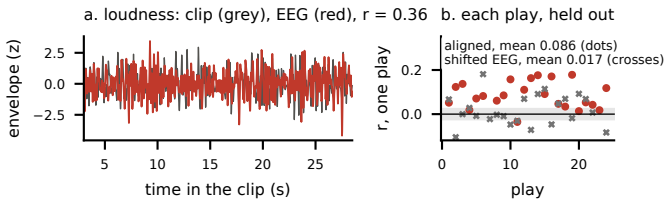


Fig. 1. The ear as a microphone. a: the loudness envelope of the 25.5 s scored from each play (grey) and the mean of the 23 held-out reconstructions from the EEG (red), $r = 0.36$. b: each play decoded by a model that never saw it, against the shift null (band) and the shifted-EEG control (crosses).

TABLE I
RHYTHM SIMILARITY OF THE SIX EXPORTED EXAMPLES TO THE HEARD CLIP (ONSET-STRENGTH CORRELATION, BEST LAG WITHIN ±60 MS). CLASSIFY-THEN-PLAY RETURNS A STORED EXAMPLE OF THE PREDICTED CLASS; FOR BOTH FAST EXAMPLES THAT FILE IS THE HEARD CLIP ITSELF.

output	fast syn.	fast str.	M2	slow syn.	slow syn. 2	slow str.	mean
EEG-conditioned	0.37	0.32	0.24	0.36	0.43	0.11	**0.31**
shuffled EEG	0.09	0.06	0.04	0.23	0.14	0.07	0.10
no EEG	0.06	0.04	0.04	0.10	0.08	0.07	0.06
classify-then-play	1.00	1.00	0.32	0.19	0.19	0.49	0.53

sit beside TP9 and TP10, so part of what comes back could be sound induced into the electrodes; the control that settles it, the same session with the headband unworn on the desk under the playing headset, was recorded twice but with too few complete plays (3 and 6) to decode, and is the first thing to repeat.

Music from the EEG. A co-worker, Ali Sher, ran the pipeline of Postolache et al. [15] on the four Muse channels: a frozen text-to-music diffusion model [16] conditioned on the EEG through a ControlNet adapter [17], the EEG denoised with the adaptive chirplet transform [18]–[20], trained on one block of eight short rhythm stimuli and tested on the other. Scored by CLAP similarity to the true stimulus [21] over about 40 trials per block, the EEG-conditioned output came closer than a model trained on shuffled EEG–stimulus pairs on both blocks (0.496 against 0.432, 0.468 against 0.418) but not significantly, while a band-power classifier named the stimulus at twice chance on one block (25%, $p = 0.023$). We scored the six exported example clips on rhythm instead: the largest correlation, within ±60 ms, between the onset-strength curves (positive spectral flux) of the heard clip and each output (Table I). The EEG-conditioned outputs followed the heard rhythm at 0.31 on average against 0.10 with shuffled EEG and 0.06 with no EEG, higher than the shuffle in all six (sign test $p = 0.016$). Six exported examples are not a held-out test, and the conditioning carries the stimulus's rhythm class, so this says the generator draws rhythm from what it is given, a step short of reading rhythm from the brain.

A steady-state attempt. The hearing test of our exam paper, after an outline by Ali Sher, tags sound with an amplitude modulation near 40 Hz and reads the ASSR at that rate behind the ears. A 1 kHz tone modulated at 40 Hz was played from the headset in three one-minute blocks at the loudest level the



Fig. 2. The ear as a hydrophone: the second author lying at the edge of the dock, the head out of the water, wearing the headband, while the hydraulophone is played alongside.

listener found comfortable. With the ear electrodes at 21 and 60 μV rms and the forehead at 8, no electrode showed a 40 Hz response in any block or pooled over the three (F at most 1.4), against a floor of 0.03 to 0.04 μV; a quieter laptop run, and one with the ear electrodes off the skin, gave nothing either, and the silent baselines gave no false alarms. The response, if present at these sites, is under about 0.1 μV; the vertex, where the ASSR is largest, has no Muse electrode, and a cup there is the next step.

IV. WATER: THE EAR AS A HYDROPHONE

The second author lay at the edge of a dock in Toronto's harbour on October 4, 2026, the head out of the water, while a hydraulophone, the water-jet instrument the second author invented, was played alongside (Fig. 2), wearing a Muse S Athena. Six minutes, no markers. The EEG was usable on 92 to 100% of 2-s epochs (under 200 μV), at 26 to 27 μV rms behind the ears and 13 to 16 on the forehead, as clean as our indoor sessions. The spectra (Fig. 3) are the ordinary 1/f shape with an alpha shoulder behind the ears; a spectrogram shows broadband bursts of movement but no narrow ridge that steps with the music, and the one steady line, at 63.9 Hz on every channel, is at the Athena's optical sampling rate and absent from recordings without the optical stream, the headband's LEDs seen by its own electrodes. A hydraulophone's fundamentals run from the low hundreds of hertz upward, all above the 128 Hz Nyquist limit, so what folds back is spread, not a line. The optical channels carried the body (Fig. 4): a pulse at 64 beats per minute on both temples, breathing near 7 per minute, and oxygenated haemoglobin rising equally on every detector pair with deoxygenated flat, a systemic change such as the posture of lying on the dock rather than a response to listening.

What a response would need. The floor behind the ears at 37 to 43 Hz was 4.5 μV^2/Hz. For an ASSR of rms amplitude A over a floor N the clean time to a signal-to-noise ratio s in one bin is $T = sN/A^2$: about one minute for 0.5 μV, three for 0.3, six for 0.2 (Fig. 5); with the head out of the water the sound must cross the water's surface, which reflects nearly all of it, so the response will be at the small end.

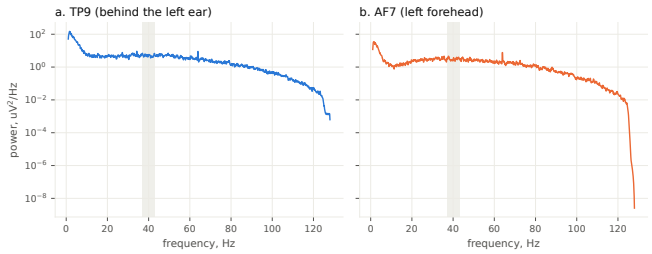


Fig. 3. Spectra of the usable epochs behind the left ear and on the left forehead at the water. The band at 37 to 43 Hz is where a 40 Hz steady-state response would appear; the line at 64 Hz is the headband's optical clock.

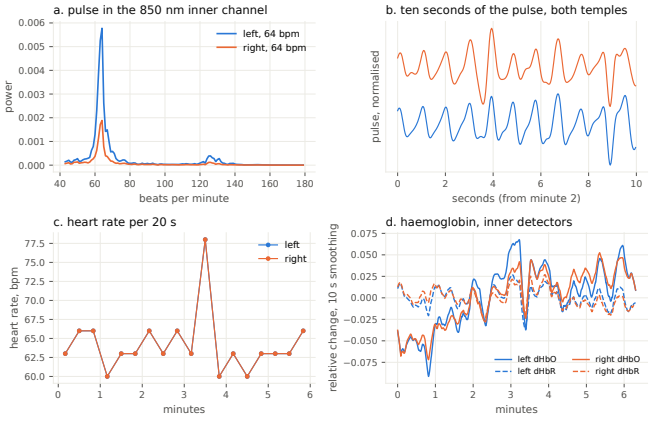


Fig. 4. The optical channels at the water. a: pulse spectrum of the 850 nm inner channel on each temple, peaking at 64 beats per minute. b: ten seconds of the pulse waveform. c: heart rate per 20 s. d: oxygenated (solid) and deoxygenated (dashed) haemoglobin on the inner detectors, relative units.

Seven listeners, one hydrophone. On October 17, 2026 the experience became a shared one (Fig. 6). A hydrophone in Lake Ontario at Love Quay fed a speaker on the pier, and a group of about seven people sat beside it, each wearing a Muse headband (Muse 1 to 7), all recorded at once by a single MuseLog phone so that every headband is on the same clock, while swimmers jumped into the lake, someone kicked the water and someone struck it with a stick. Listening together is where a wearable that reads many people at once earns its place: group EEG has tracked shared attention in a classroom [22], and the inter-subject correlation of brain activity rises when people take in the same natural stimulus [23]. A shared sound in the water should, if the headbands can see it, make several listeners react in the same second.

Four group sessions were recorded after a short test: at 15:44 (55 s, six headbands with data), 15:48 (92 s, six), 15:53 (208 s, six) and 16:09 (217 s, four). The phone logged no event markers, so the analysis looks for reactions the listeners share rather than responses to known moments. For each headband we took the frontal EEG (AF7 and AF8, 0.5 to 8 Hz, where blinks and startle appear) and the head's rotation from the gyroscope in 1-s bins, scored each against its own session (robust z), and counted the seconds in which at least two

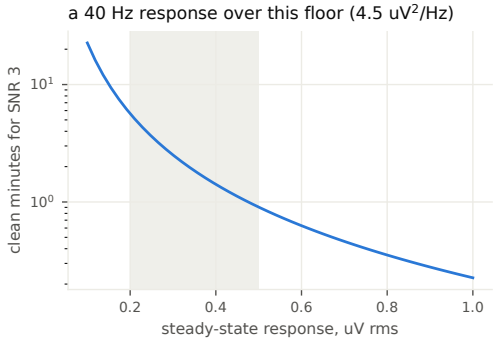


Fig. 5. Clean recording time for a 40 Hz steady-state response to reach a signal-to-noise ratio of three over the floor measured behind the ears at the water, against the response's amplitude.

listeners exceeded $z = 2.5$ together, against 1,000 surrogates that shift each listener's series by an independent random lag; the mean pairwise correlation between listeners (ISC) was tested the same way. These settings were fixed before the counts were seen.

No session showed a shared moment beyond chance (Fig. 7): 0, 6, 1 and 2 shared seconds in the frontal EEG against 0.4, 4.3, 0.9 and 2.1 expected (p 0.25 to 1), and the most listeners reacting in any one second was at most two, which chance gives somewhere in a session 63 to 100% of the time. The ISC of the frontal EEG was 0.02 to 0.09, above its surrogates only at 15:48 ($r = 0.085$, $p = 0.02$), which does not survive four sessions. Head rotation could be read on only one or two headbands per session and showed nothing shared either. Three things stand between this afternoon and an answer: the phone in this app version logged a constant contact value per headband, so contact could not be checked beyond the signal itself; the sounds were not timed; and the hydrophone's sound reached the listeners through the air from a speaker, after crossing the water's surface. What it shows is that seven consumer headbands can be put on a group at the water's edge and recorded on one clock in minutes, which is the instrument for the question, if not yet the answer.

V. Solid: The Ear as a Geophone

On November 4, 2026 the first author sat at a table, the head not touching it, and made sound in it, recorded by MuseLog on the phone: two minutes of tapping the table with the fingernails, one of scratching it, and one of tapping it with the fingers, 4.1 minutes in all. The headband rated all four electrodes in good contact throughout (contact index 1 on every channel). Three questions follow the solid path into the head (Fig. 8).

Did the head move with the gestures? Yes. The accelerometer, high-passed at 3 Hz, shows the head moving at 3.8 mg rms during the fingernail taps, 2.8 during scratching and 5.6 during the finger taps, with broad spectra peaking at 6.6, 4.6 and 4.9 Hz. With the head off the table this is the gesture

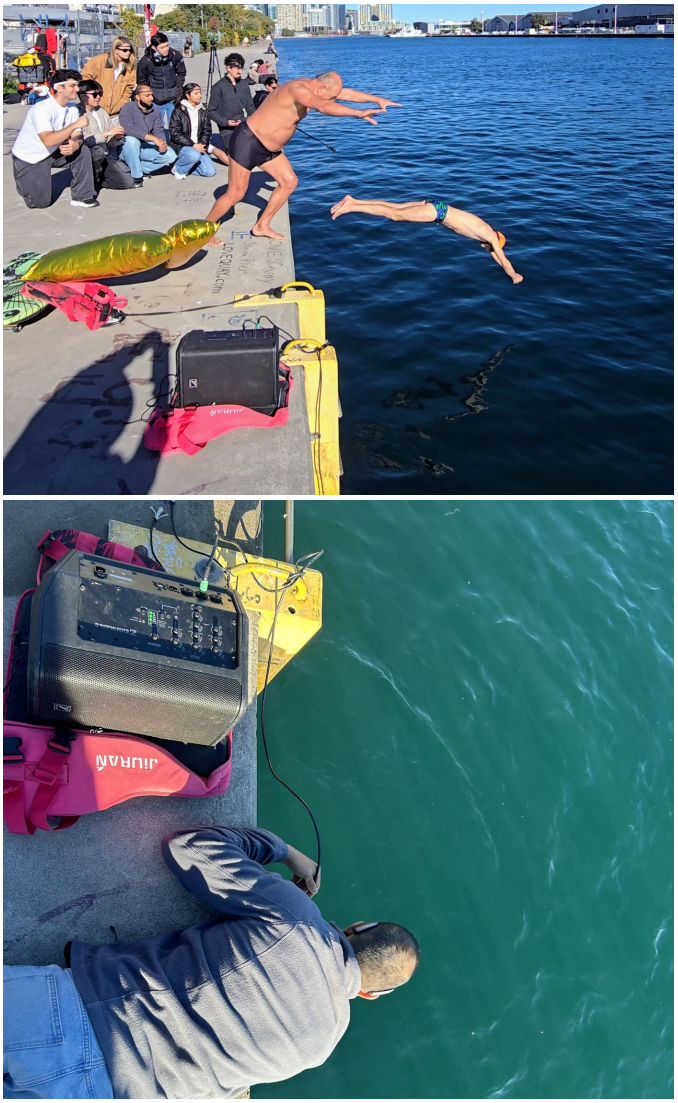


Fig. 6. Seven listeners, one hydrophone, Love Quay, October 17, 2026. Top: the group, wearing Muse headbands, listens to the hydrophone's speaker on the pier as swimmers jump into Lake Ontario. Bottom: striking the water with a stick beside the hydrophone's cable, the speaker on the pier.

carried through the arms and body, not the table's vibration; the sound itself reached the ear through the air.

Did the EEG follow it? No. The coherence between the head's motion and each electrode at that peak was 0.00 to 0.12, at or below each electrode's own median coherence from 1 to 15 Hz, in every segment; the 40 to 120 Hz power behind the ears was the same in all three (4.2 to 4.5 μV^2/Hz on TP9), so scratching left no broadband trace either.

Was there a response to each tap? Not reliably. Taps found as the head motion's largest peaks (22 usable epochs for the fingernails, 34 for the fingers) gave tap-locked averages behind the ears inside the band of averages at random times for the finger taps (7% of samples outside, chance 5%); the fingernail average left the band on 20% of samples, with a positive deflection of 4.5 μV near 120 ms, but it left it as far before the tap as after, so it is not a response. With taps timed only to the accelerometer's 39 Hz samples, an auditory evoked potential would also be smeared by up to 25 ms.

VI. One Protocol for Three Media

The three recordings agree on one lesson: what the headband hears of a sound is what was tagged and timed. The speech envelope came back because the same clip was played 24 times at known moments; the music, the hydraulophone and the table were not tagged, and gave the body's response (pulse, posture, motion) rather than the brain's. The protocol that follows is the same in every medium. *A tagged source:* a speaker in air, a transducer in the water or a hydraulophone jet, a shaker on the table, each driven with an amplitude modulation at a rate the headband samples well, 40 Hz first, in 8-s on and off blocks, twenty of each, with the block edges logged on the recording's clock; then an exact permutation test over the block labels, as our visual calibration does for flicker. *Event timing for impulses:* a contact microphone or the headset's own audio clock for each tap, not the accelerometer, so an evoked potential can be averaged. *A cup at the vertex,* where auditory responses are largest and the Muse has no electrode. *A desk control* with the headband unworn beside the source, for every medium, so that induction and vibration of the electrodes are measured rather than assumed away. *Timed events for a group:* the hydrophone's own audio recorded on the phone's clock beside the headbands, so that every splash, kick and strike becomes a marker and the group's responses can be averaged across people, with contact logged per second [24]. *The path as the contrast:* the same tagged source in the water or the table heard with the head out of it, as here, and then with the ear against it, so that crossing the boundary and bone conduction are measured apart, in one sitting.

Limits. One listener per medium, one session each, no markers at the water or on the table, and the air-path control not yet decodable. Nothing here is a measurement of hearing thresholds in water or through solids; it is what a consumer headband recorded, and what it would take to record more.

VII. Conclusion

Through air, the ear is a microphone on a consumer headband: a spoken passage's loudness comes back from four dry electrodes and a cup at $r = 0.36$, and an EEG-conditioned generator borrows the heard rhythm. Seven listeners at the water's edge can be recorded on one clock in minutes, though their headbands did not yet share a reaction to the splash. Through water and through a solid, the headband records the body, a pulse at 64 beats per minute at the dock and the head's motion at the table, but not yet the brain's hearing of either. The step that made the eye a camera, a tagged stimulus with logged blocks and a permutation test, is the step that will make the ear a hydrophone and a geophone, and the protocol is set out above.

Acknowledgements

We thank Ali Sher (Mersivity Lab, University of Toronto) for the EEG-to-music pipeline and its clips and for the

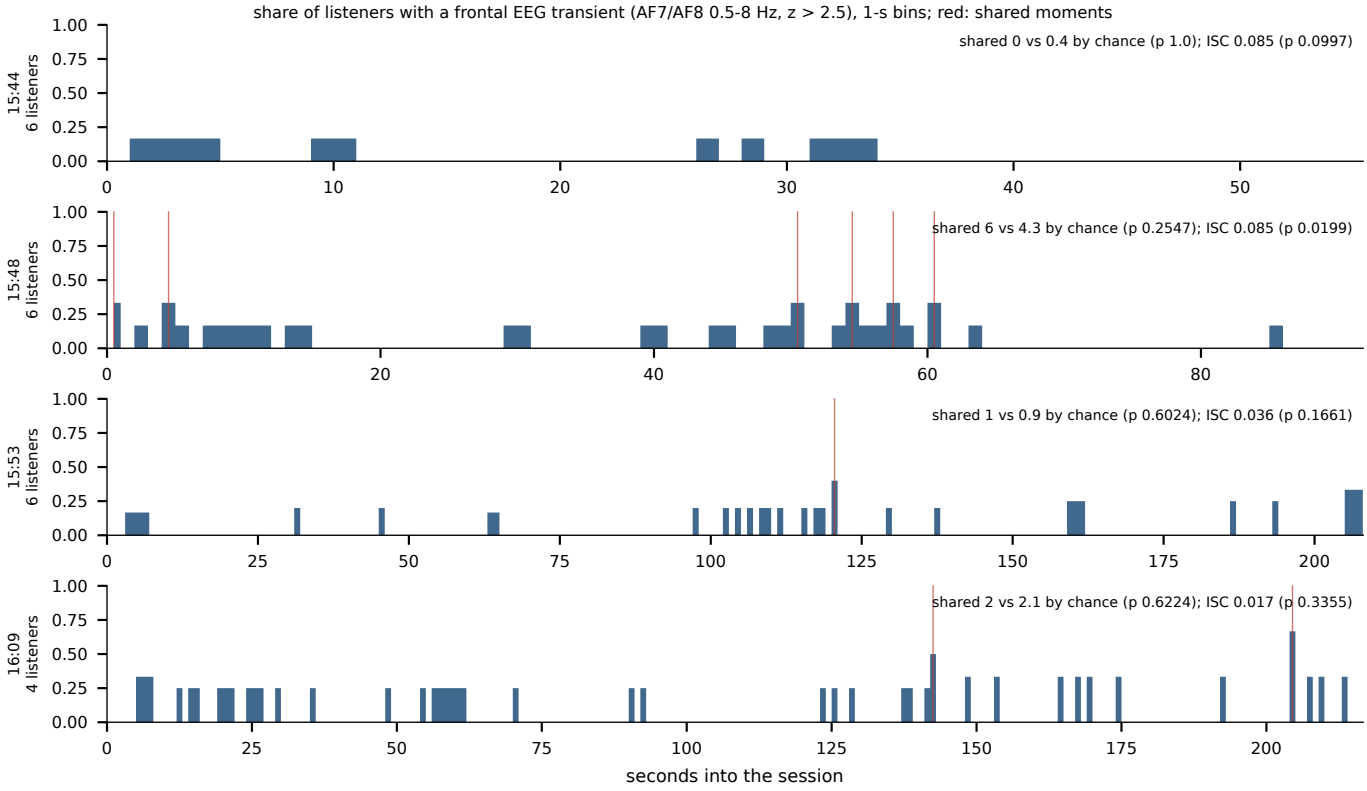


Fig. 7. The four group sessions on one clock. For each second, the share of listeners whose frontal EEG (AF7/AF8, 0.5 to 8 Hz) exceeded $z = 2.5$; red lines mark seconds in which at least two did. Each panel gives the shared seconds against the number expected from 1,000 independently shifted surrogates, and the inter-subject correlation with its surrogate p.

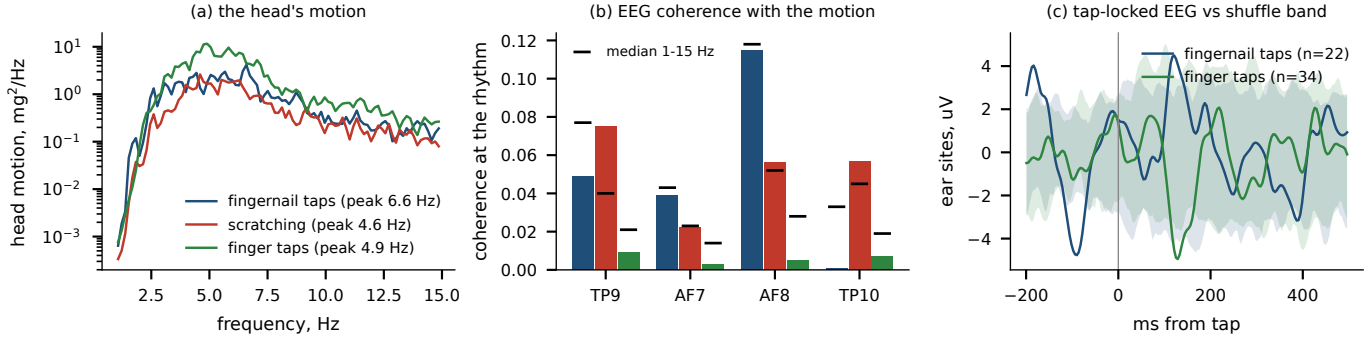


Fig. 8. The ear as a geophone: tapping and scratching a table, the head not touching it. (a) The head's motion from the headband's accelerometer (3 Hz high-pass) in each segment. (b) Coherence between that motion and each electrode at the segment's peak, against each electrode's median coherence from 1 to 15 Hz (black ticks). (c) EEG behind the ears averaged around the taps, against the 95% band of averages at random times in the same segment.

hearing-test outline, the seven listeners of October 17 and the swimmers, the hydraulophone players and the dock crew, MannLab Canada, and Interaxon.

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